Kaduna.lk – A Unified Smart Platform for Roadside Assistance and Automotive Services in Sri Lanka

R26-SE-026

Project Proposal Report

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Component: Geo-Intelligence & Traffic Impact Analysis

B.Sc. (Hons) Degree in Information Technology Specialized in Software Engineering

Department of Information Technology

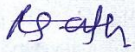
Sri Lanka Institute of Information Technology

Sri Lanka

March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Sri Lanka's roadside assistance ecosystem is fragmented and manually coordinated, resulting in delayed incident clearance, prolonged traffic congestion, and misallocated recovery resources. When a vehicle breaks down or an accident occurs, there is no mechanism to assess the traffic severity of the incident, causing all breakdowns to be treated with equal urgency regardless of their actual impact on road network performance. Research shows that the relationship between incident duration and congestion is non-linear—reducing incident duration by 50% can cut total traffic delays by up to 75% [1]. Yet no published study has quantified this relationship for Sri Lankan road conditions. This research addresses this gap by proposing a Geo-Intelligence and Traffic Impact Analysis engine as part of the Kaduna.lk unified smart platform. The proposed component introduces an incident impact scoring model that will calculate a severity score (1–10) based on five weighted factors: capacity loss, traffic volume, temporal conditions, location characteristics, and incident severity. The model will be validated against SUMO microscopic traffic simulations, targeting a Pearson correlation of $r > 0.75$ between predicted scores and simulated congestion metrics. Additionally, spatial hotspot analysis using Kernel Density Estimation (KDE) and DBSCAN clustering will identify high-risk breakdown zones across the Colombo road network. An interactive visualization dashboard built with Next.js and Leaflet.js will present real-time incident intelligence, congestion predictions, and a what-if simulation tool. The key technologies to be employed include Python, SUMO traffic simulator, OSMnx, scikit-learn, SciPy, and Next.js with Leaflet.js. Expected outcomes include a validated impact scoring model for Sri Lankan roads, quantified congestion reduction through priority-based dispatch, and identified spatial patterns of vehicle incidents. The social impact includes reduced traffic congestion and improved emergency response, while the commercial impact lies in enabling smarter resource allocation for roadside assistance providers.

Keywords: Traffic Impact Scoring, Geo-Spatial Analysis, Incident-Induced Congestion, SUMO Traffic Simulation, Roadside Assistance Optimization

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my project supervisor, Ms. Karthiga Rajendran, Department of Software Engineering, Sri Lanka Institute of Information Technology, for her continuous guidance, constructive feedback, and encouragement throughout this research. Her expertise in software systems and research methodology has been invaluable in shaping this work.

I extend my appreciation to the Department of Information Technology and the Faculty of Computing at SLIIT for providing the academic environment and resources necessary for this research project.

I am grateful to my team members—Janukshan S, Herath D M S T, and De Silva R K D H—for their collaboration on the Kaduna.lk platform. Their work on dispatch optimization, predictive maintenance, and compliance respectively has provided the broader context within which this geo-intelligence component operates.

Finally, I thank my family and friends for their patience and support throughout this academic journey.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
CLF	Capacity Loss Factor
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
GIS	Geographic Information System
KDE	Kernel Density Estimation
ISF	Incident Severity Factor
LF	Location Factor
LWR	Lighthill-Whitham-Richards (traffic flow model)
OSM	OpenStreetMap
PDPA	Personal Data Protection Act
SDG	Sustainable Development Goal
SUMO	Simulation of Urban Mobility
TF	Temporal Factor
TVF	Traffic Volume Factor
VHL	Vehicle-Hours Lost

1. INTRODUCTION

1.1 Background and Context

Sri Lanka's transportation ecosystem faces significant challenges during vehicle breakdowns and road accidents. When a vehicle breaks down or a collision occurs, roadside assistance is typically arranged through manual phone calls to fragmented service providers, resulting in delayed response times, increased driver stress, and prolonged road blockages. These challenges disproportionately affect daily commuters, commercial vehicle operators, elderly drivers, and women drivers, while also impacting traffic authorities and emergency responders.

The impact of delayed incident clearance extends far beyond the individual driver. When a vehicle blocks a lane on a busy road, the remaining capacity is reduced, causing a shockwave of congestion that propagates backward from the incident. On Sri Lanka's urban roads—particularly in Colombo, Kandy, and Galle—which already operate near capacity during peak hours, a single blocked lane on a two-lane road can reduce capacity by 50% or more. With limited alternative routes in many areas, the impact of any blockage is amplified significantly.

Research presented at the Transport Research Arena Conference demonstrated that the relationship between incident duration and congestion is non-linear: reducing incident duration by 50% can cut total traffic delays by up to 75% [1]. This finding has profound implications—intelligently prioritizing which incidents receive the fastest response can yield disproportionate benefits for the entire road network. Yet current roadside assistance systems treat all incidents identically, with no mechanism to assess, predict, or prioritize based on traffic impact.

Existing digital solutions in Sri Lanka's mobility sector are fragmented and lack integration. There is no single platform unifying roadside assistance, garages, and service providers. Drivers must rely on multiple phone calls or apps for towing, repairs, or emergency services. Critically, most local mobility applications have not fully adopted Sri Lanka's Personal Data Protection Act (PDPA) requirements, raising privacy concerns and eroding user trust.

The Kaduna.lk platform is proposed as a unified, smart, and PDPA-compliant solution to address these challenges. The platform comprises four integrated components: (1) Dispatch Optimization and Smart Recovery, (2) Geo-Intelligence and Traffic Impact Analysis, (3) Predictive Maintenance for Service Management, and (4) Compliance, Integration, and Stakeholder Engagement. This proposal focuses specifically on Component 2—the Geo-Intelligence and Traffic Impact Analysis engine.

This research aligns with the SST – Software Systems and Technologies research cluster and the Transportation / Smart Mobility industry vertical. It supports United Nations Sustainable Development Goals including SDG 9 (Industry, Innovation, and Infrastructure), SDG 11 (Sustainable Cities and Communities), and SDG 13 (Climate Action) by reducing unnecessary congestion, fuel wastage, and emissions caused by delayed incident clearance.

1.2 Literature Review

A comprehensive review of existing literature was conducted across academic databases (IEEE, Springer, MDPI, Transportation Research Record), industry standards, and open-source tool documentation to identify the current state of research in incident-induced traffic congestion, traffic impact assessment, spatial hotspot analysis, and congestion prediction. The review is organized into four thematic areas directly relevant to the proposed geo-intelligence engine.

1.2.1 Traffic Impact of Vehicle Incidents and Congestion Modelling

The impact of vehicle breakdowns on motorway congestion was investigated using the VISSIM microscopic traffic simulator with real incident data from the M1 motorway in the UK [1]. The study simulated various Connected Vehicle (CV) market penetration scenarios and found that at 100% CV adoption, average vehicle delays decreased by 50% and conflicts were reduced by 55–70%. This is the most directly relevant study to the current research, demonstrating the non-linear relationship between incident detection speed and congestion severity. However, the work was limited to UK motorway conditions with well-defined lane structures and did not develop a scoring model or prioritization framework applicable to mixed-traffic conditions in developing countries [1].

The foundational theory of traffic flow on congested roads was established through the kinematic wave model (LWR model), which describes how shockwaves propagate through traffic when capacity is reduced [2]. When a lane is blocked, vehicles arrive faster than they can pass through the remaining capacity, creating a backward-propagating shockwave whose growth rate depends on the difference between arrival rate and reduced capacity. This theoretical framework underpins the congestion prediction component of the current research, providing the mathematical basis for estimating queue length, vehicle-hours lost, and recovery time after incident clearance [2].

The Highway Capacity Manual (HCM), 7th Edition, provides standardized methodologies for calculating road capacity, level of service, and the impact of incidents on traffic flow. The HCM defines capacity reduction factors for various incident types: a single-lane closure on a two-lane road reduces capacity by approximately 50%, while on a four-lane road the reduction is approximately 25%. These capacity values inform the Capacity Loss Factor and Traffic Volume Factor calculations in the proposed impact scoring model [3].

The severity of road accident impact on traffic flow was identified using an ensemble model combining Random Forest and Gradient Boosting classifiers [4]. The study classified incidents into severity levels based on traffic flow disruption metrics and demonstrated that ensemble methods outperform single classifiers for incident severity prediction. While the approach focused on post-hoc classification rather than real-time scoring, it validates the feasibility of multi-factor severity assessment—a core principle of the proposed impact scoring model [4].

1.2.2 Traffic Simulation and Incident Modelling

A SUMO-based evaluation of road incidents' impact on traffic congestion levels in smart cities extended the simulator with a real-time vehicle re-routing mechanism that allows vehicles to bypass blocked roads during simulation runtime [5]. Results demonstrated significant efficiency improvements in reducing travel time during incidents compared to basic SUMO. This work validates SUMO as a suitable platform for incident-induced congestion analysis and informs the simulation methodology of the current research [5].

A Python extension for SUMO was developed to simulate traffic incidents and emergency service vehicle response [6]. The tool enables random traffic incident generation with configurable location and duration, incident detection capabilities, and emergency vehicle dispatching. This extension addresses a critical gap in testing traffic management systems under incident conditions and produces training data for reinforcement learning models. The current research adapts similar SUMO incident simulation techniques using variableSpeedSign elements for lane closure modelling [6].

Extreme Gradient Boosting (XGBoost) was applied to predict traffic incident clearance time and analyse influencing factors [7]. The study found that incident type, number of lanes blocked, and time of day were the most significant predictors of clearance duration. The non-linear relationship between these factors and clearance time supports the multi-factor weighted approach adopted in the proposed impact scoring model [7].

1.2.3 Spatial Hotspot Analysis and GIS Methods

The DBSCAN (Density-Based Spatial Clustering of Applications with Noise) algorithm was proposed for discovering clusters of arbitrary shape in spatial databases [8]. Unlike K-means, DBSCAN does not require specifying the number of clusters in advance and can identify noise points. The algorithm groups points based on two parameters: epsilon (maximum distance) and min_samples (minimum points to form a cluster). This makes it particularly suitable for identifying incident hotspot zones where breakdowns cluster spatially without assuming regular-shaped clusters [8].

Kernel Density Estimation (KDE) bandwidth was optimized for road traffic accident hazard identification in the City of London [9]. The study used incremental spatial autocorrelation analysis to determine an optimal bandwidth of 134 meters, achieving a predictive accuracy index of 4.381. This work demonstrates that bandwidth selection critically affects hotspot detection quality—a consideration adopted in the current research's KDE implementation for Colombo incident data [9].

Getis-Ord Gi* statistics were combined with DBSCAN to identify urban traffic accident hotspots in an IEEE-published study [10]. The dual-method approach demonstrated that combining global spatial autocorrelation with density-based clustering produces more robust hotspot identification than either method alone. The current research similarly employs complementary methods (KDE for continuous density surfaces and DBSCAN for discrete cluster identification) [10].

1.2.4 Road Network Analysis and Incident Management Tools

The official OSMnx reference paper describes the Python package's capabilities for downloading, modelling, and analysing urban street networks from OpenStreetMap data [11]. OSMnx enables programmatic access to road network data with attributes including road type, lane count, speed limits, and geometry. It has become the standard tool for computational street network analysis in urban science research. For this research, OSMnx is used to download the complete Colombo road network—essential input for the geo-intelligence engine [11].

Roadside assistance dispatch was formulated as a multi-objective optimization problem for a Spanish automobile club, using heuristics and linear programming considering skill matching, priorities, and cost [12]. The approach automated 100% of incident assignments (up from 40%) and achieved a 19% reduction in operational costs. While this demonstrates the potential of algorithm-driven dispatch, it does not incorporate traffic impact intelligence into prioritization—all incidents are prioritized based on contractual SLAs rather than their actual impact on road network performance [12].

A foundational real-time decision support system for roadway network incident response logistics was developed to minimize incident response time through districting, dispatching, and resource allocation optimization [13]. This work established the theoretical basis for integrating traffic intelligence with dispatch decisions—a principle central to the current research's approach of using impact scores to inform dispatch priority [13].

1.2.5 Summary of Related Work

Domain	Source	Approach	Key Finding	Limitation
Traffic Impact	Koliou et al. (2024) [1]	VISSIM simulation, V2V	50% delay reduction at 100% CV	UK motorways only; no scoring model
Traffic Theory	Lighthill & Whitham (1955) [2]	LWR kinematic wave model	Foundational shockwave theory	Needs calibration for local conditions
Capacity	HCM 7th Ed. (2022) [3]	Standardized capacity methods	Incident capacity reduction factors	Western road standards; needs adaptation
Severity	Li et al. (2021) [4]	Ensemble ML classification	Multi-factor severity assessment	Post-hoc, not real-time scoring
Simulation	Abubakar et al. (2023) [5]	SUMO incident simulation	Re-routing reduces travel time	Limited incident types tested
Simulation	Bieker-Walz et al. (2023) [6]	SUMO Python extension	Incident + emergency vehicle sim	Focused on signal control testing
Clearance	Tang et al. (2020) [7]	XGBoost prediction	Lanes blocked + time are key factors	Clearance time, not impact scoring
Clustering	Ester et al. (1996) [8]	DBSCAN algorithm	Arbitrary-shape cluster discovery	Requires parameter tuning
KDE	Ouni & Belloumi (2024) [9]	Optimized KDE bandwidth	PAI = 4.381 for London accidents	Single-city study
Hotspots	Kaygisiz et al. (2024) [10]	Getis-Ord Gi* + DBSCAN	Dual-method robust identification	No impact severity weighting
GIS Tools	Boeing (2025) [11]	OSMnx Python library	Standard for street network analysis	Data quality varies by region

Dispatch	Buil et al. (2021) [12]	Multi-objective optimization	100% automated, 19% cost reduction	No traffic impact consideration
Incident Mgmt	Ozbay & Kachroo (1999) [13]	Real-time DSS	Integrated dispatch optimization	Pre-dates modern ML methods

1.3 Research Gap

Despite extensive work on traffic simulation, dispatch optimization, and predictive maintenance globally, several critical gaps remain—particularly in the Sri Lankan context—when it comes to connecting traffic impact intelligence with roadside assistance operations. The literature review reveals the following gaps:

Gap 1: No incident impact scoring model for Sri Lankan roads

While traffic simulation studies exist for UK and European motorways [1], and ensemble severity classification has been demonstrated [4], no published research has developed a real-time traffic impact scoring model calibrated for Sri Lankan road network characteristics. Sri Lankan roads present unique challenges: 73% of roads with lane data are two-lane roads (high vulnerability to single-lane blockages), 78% are classified as residential (narrow, with limited alternative routes), and urban roads in Colombo, Kandy, and Galle already operate near capacity during peak hours. These conditions create different congestion dynamics than the well-structured motorway networks studied in Western literature. A locally calibrated scoring model is needed to accurately predict incident severity in this context.

Gap 2: Dispatch systems lack traffic intelligence

Existing dispatch optimization research [12] focuses on resource allocation efficiency—matching providers to incidents based on proximity, skills, and cost—but does not incorporate traffic impact assessment into the prioritization process. Current systems assign providers based on distance alone, treating a breakdown on an empty rural road identically to one blocking the Colombo-Katunayake Expressway at rush hour. Even foundational incident management DSS work [13] did not incorporate congestion severity scoring into dispatch logic. This means that a tow truck might be dispatched to a low-impact rural breakdown while a high-impact expressway incident causes cascading congestion.

Gap 3: No spatial analysis of vehicle incidents in Sri Lanka

No published hotspot analysis of vehicle breakdowns or accidents exists for Sri Lankan roads. While spatial clustering techniques such as DBSCAN [8] and Kernel Density Estimation [9] are well-established in geographic analysis—and have been applied successfully in London [9] and other cities [10]—they have not been applied to identify high-risk breakdown zones in Sri Lanka. Furthermore, existing hotspot analyses focus only on incident frequency, not on the combined metric of frequency and traffic impact severity. The proposed composite risk score (Incident Count × Average Impact Score) addresses this limitation.

Gap 4: Separation of traffic analysis and dispatch optimization

Current literature treats traffic analysis and dispatch optimization as separate research problems. Traffic engineers study congestion patterns using simulation [5][6]; operations

researchers optimize resource allocation [12]. This research bridges the two domains by using traffic intelligence to inform dispatch decisions, enabling impact-aware prioritization. The non-linear relationship between clearance time and congestion [1][7] means that intelligently prioritizing which incidents get the fastest response can yield disproportionate benefits for the entire road network—a finding that has not been operationalized in any existing roadside assistance platform.

Gap 5: Lack of predictive congestion tools for Sri Lankan traffic authorities

Sri Lankan traffic authorities currently learn about congestion after it has already formed. There is no tool that predicts future congestion from a newly reported incident using shockwave theory [2] calibrated for local road conditions [3]. Google Maps shows current traffic conditions—what is happening right now—but cannot predict that a breakdown reported 30 seconds ago will cause 2 km of queuing within 15 minutes. The proposed geo-intelligence engine fills this gap by providing predictive intelligence: estimating congestion spread before it develops, enabling proactive traffic management rather than reactive observation.

2. OBJECTIVE

2.1 Main Objective

To design and develop a geo-intelligence engine that quantifies the traffic impact of vehicle incidents on Sri Lankan roads through an incident impact scoring model, spatial hotspot analysis, and traffic simulation, enabling priority-based dispatch for maximum congestion reduction.

2.2 Specific Objectives

SO1: To develop a multi-factor incident impact scoring model (1–10 scale) incorporating capacity loss, traffic volume, temporal conditions, location characteristics, and incident severity for Sri Lankan road networks.

SO2: To validate the scoring model against SUMO microscopic traffic simulation results, achieving a Pearson correlation coefficient of $r > 0.75$ between predicted impact scores and simulated congestion metrics.

SO3: To identify spatial clusters (hotspots) of vehicle incidents in the Colombo metropolitan area using Kernel Density Estimation and DBSCAN clustering, and correlate hotspot locations with road network characteristics.

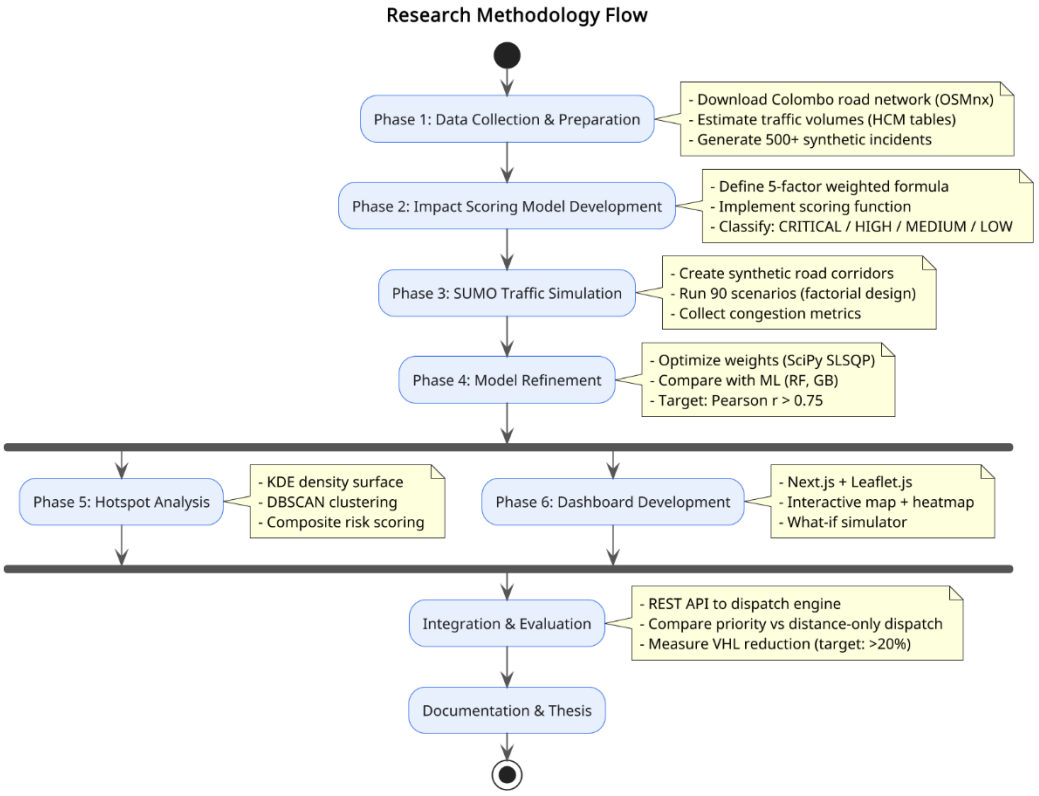
SO4: To develop a congestion prediction model based on Lighthill-Whitham-Richards shockwave theory that estimates queue length, vehicle-hours lost, and recovery time for scored incidents.

SO5: To build an interactive visualization dashboard with a what-if simulation tool that presents incident intelligence, congestion predictions, and hotspot maps on an interactive web-based map.

SO6: To demonstrate that priority-based dispatch (using impact scores) reduces total network vehicle-hours lost by at least 20% compared to distance-only dispatch through simulation scenario comparison.

3. METHODOLOGY

This research follows a Design Science Research approach combined with experimental simulation methodology. The research is structured into six phases, each producing verifiable outputs that feed into subsequent phases. The overall methodology flow is illustrated in the figure below.



Research Questions

RQ1: Can a geo-spatial incident impact scoring model accurately predict congestion severity based on road characteristics, traffic conditions, and temporal factors in Sri Lankan road networks?

RQ2: How can real-time traffic intelligence be integrated with dispatch systems to prioritize incidents that will cause the greatest traffic disruption?

RQ3: What are the spatial patterns of vehicle incidents in urban Sri Lanka, and how do road characteristics (lane count, road type, proximity to junctions) correlate with incident-induced congestion severity?

Hypotheses

H1: An incident impact scoring model incorporating road type, lane count, time of day, and baseline traffic volume can predict congestion severity with at least 75% accuracy (Pearson r > 0.75) when validated against SUMO simulation ground truth.

H2: Priority-based dispatch using impact scores reduces total network vehicle-hours lost by at least 20% compared to distance-only dispatch.

H3: Vehicle incidents in the Colombo metropolitan area exhibit statistically significant spatial clustering, with identifiable hotspot zones that correlate with road network characteristics.

Methodology Phases

Phase 1: Data Collection and Preparation

Road network data for the Colombo metropolitan area will be obtained from OpenStreetMap using the OSMnx Python library [11]. The network is expected to contain tens of thousands of nodes and edges covering the full road network of the study area. Each edge will carry attributes including road type (trunk, primary, secondary, tertiary, residential), lane count, speed limit, and geometry. Preliminary analysis of Sri Lankan road networks suggests that the majority of roads are classified as residential with two-lane configurations—indicating high vulnerability to single-lane blockages.

Traffic volume data will be estimated using road classification-based capacity tables adapted from the Highway Capacity Manual (HCM) [3] for Sri Lankan conditions. Where available, Google Maps Traffic API data will supplement these estimates. Incident data will be generated synthetically using a Poisson distribution for frequency, spatial distribution weighted by road type and traffic volume, and temporal distribution matching known peak patterns. A target dataset of at least 500 incident records will be generated for the Colombo district, each with location (GPS coordinates), incident type, lanes blocked, time of occurrence, and road segment attributes. This synthetic data generation approach is consistent with accepted methodology in transportation research [1]. If real incident data becomes available from police reports or insurance records, it will be incorporated to improve the analysis.

Phase 2: Impact Scoring Model Development

The incident impact scoring model will compute a severity score (1–10) using a weighted combination of five factors:

$$\text{Impact_Score} = (w_1 \times \text{CLF} + w_2 \times \text{TVF} + w_3 \times \text{TF} + w_4 \times \text{LF} + w_5 \times \text{ISF}) \times 10$$

Factor	Calculation	Description
Capacity Loss Factor (CLF)	lanes_blocked / total_lanes	Fraction of road capacity removed
Traffic Volume Factor (TVF)	current_volume / road_capacity	How close to saturation the road is
Temporal Factor (TF)	Time-based multiplier (0.1–1.0)	Rush hour vs. off-peak vs. night
Location Factor (LF)	Road classification multiplier	Expressway > A-road > residential
Incident Severity Factor (ISF)	Incident type multiplier (0.2–1.0)	Major accident > engine failure > flat tire

Initial weights will be assigned based on domain knowledge and literature review (CLF=0.25, TVF=0.25, TF=0.20, LF=0.15, ISF=0.15) and subsequently optimized using SUMO simulation results through sensitivity analysis and correlation maximization. The score will be normalized to a 1–10 scale and classified into four priority levels: CRITICAL (8.0–10.0), HIGH (5.0–7.9), MEDIUM (3.0–4.9), and LOW (1.0–2.9).

The initial weight distribution reflects the relative importance of each factor as established in the transportation engineering literature. Capacity Loss Factor and Traffic Volume Factor receive the highest weights (0.25 each) because empirical studies have consistently identified lane closures and traffic demand as the dominant determinants of incident-induced delay — Garib et al. (1997) demonstrated that incident duration, number of lanes affected, and traffic demand explain up to 85% of variation in incident delay on urban freeways. The Temporal Factor receives a moderate weight (0.20) as time-of-day effects were identified as a significant predictor by both Giuliano (1989) and Tang et al. [7]. Location Factor and Incident Severity Factor receive lower initial weights (0.15 each) as their effects are partially captured through the other factors. Critically, these initial weights serve only as a starting point — they are subsequently optimized using SUMO simulation data through correlation maximization (see Phase 4), ensuring the final model is empirically calibrated rather than assumption-dependent.

Phase 3: SUMO Traffic Simulation and Validation

The SUMO (Simulation of Urban Mobility) microscopic traffic simulator [5][6] will be used to generate ground-truth congestion data for model validation. Synthetic road corridors will be created for each road type, with lane closures simulated using SUMO's variableSpeedSign or rerouter elements to model incident-induced capacity reduction. Edge-level traffic data will be collected at regular intervals throughout each simulation run.

A factorial experiment design will produce approximately 90 scenarios across five dimensions:

Dimension	Values	Count
Road types	trunk, primary, secondary, tertiary, residential	5
Lane configurations	2 lanes, 3 lanes	2
Volume fractions	30%, 60%, 90% of capacity	3
Incident durations	15 min, 30 min, 60 min	3
Total scenarios		90

For each scenario, congestion metrics will be recorded: speed reduction percentage, vehicle-hours lost, and congestion severity index. These metrics will serve as ground truth for validating the impact scoring model. The target is to achieve a Pearson correlation of $r > 0.75$ between predicted impact scores and simulated congestion severity.

Phase 4: Model Refinement and Weight Optimization

Weight optimization will be performed using scipy.optimize.minimize with the SLSQP method, maximizing the Pearson correlation between formula-based impact scores and SUMO-measured speed reduction. The optimization will be constrained such that all weights are non-negative and sum to 1.0. This approach will reveal which factors are the dominant predictors of congestion severity—we hypothesize that Capacity Loss Factor and Location

Factor will emerge as the most influential, which would be physically intuitive (blocking more lanes on more important roads should cause more congestion).

A comparison with machine learning approaches (Random Forest, Gradient Boosting) will also be conducted to determine whether data-driven models outperform the domain knowledge-based formula. Cross-validation will be used to assess generalization performance and detect overfitting. If ML models show superior cross-validated performance, they may replace the formula-based approach; otherwise, the interpretable formula model will be preferred for its transparency and explainability.

Phase 5: Hotspot Analysis

Spatial clustering will be performed on incident locations using two complementary methods: (1) Kernel Density Estimation (KDE) using scipy.stats.gaussian_kde to create a continuous density surface showing where incidents concentrate, and (2) DBSCAN clustering [8] with haversine distance metric to identify discrete hotspot zones. DBSCAN parameters (epsilon and min_samples) will be tuned based on the spatial distribution of the study area. A composite risk score will be calculated as Incident Count × Average Impact Score, ranking hotspots by both frequency and severity rather than incident count alone. This impact-weighted approach is a key contribution—a location with fewer incidents but higher average impact may be more important than a high-frequency, low-impact location.

Phase 6: Dashboard Development and Integration

An interactive visualization dashboard will be developed using Next.js (App Router, TypeScript) with Tailwind CSS for styling and Leaflet.js with react-leaflet for map rendering. The planned dashboard features include: an interactive map with color-coded incident markers by severity, a density heatmap weighted by impact score, hotspot zone overlays showing DBSCAN cluster boundaries, priority and road type filters, a statistics panel with KPIs and distribution charts, an incident detail panel showing score breakdown and congestion predictions, and a what-if simulator that allows users to adjust parameters (road type, lanes blocked, time of day, incident type) and see predicted impact scores in real-time. The dashboard will serve as the primary prototype deliverable for demonstrating the geo-intelligence engine's capabilities.

Evaluation Metrics and Validation Strategy

The following metrics will be used to evaluate the performance of each component of the geo-intelligence engine:

Component	Metric	Target	Description
Impact Scoring Model	Pearson correlation (r)	> 0.75	Correlation between predicted scores and SUMO-measured congestion
Impact Scoring Model	Mean Absolute Error (MAE)	< 1.5 score points	Average deviation between predicted and simulated severity

Congestion Prediction	Queue length error	< 20%	Difference between predicted and simulated queue length
Congestion Prediction	VHL prediction error	< 25%	Difference between predicted and simulated vehicle-hours lost
Hotspot Analysis	Silhouette score	> 0.3	Cluster quality measure for DBSCAN output
Hotspot Analysis	Spatial coverage	> 80% of high-impact incidents	Proportion of high-impact incidents captured by hotspot zones
Priority Dispatch	VHL reduction	> 20%	Reduction in total network VHL with priority vs. distance-only dispatch
Dashboard	Load time	< 3 seconds	Time to render map with 50+ incidents
Scoring API	Response time	< 500ms	Time to compute and return impact score for a single incident

Tools and Technologies

Tool	Purpose	Usage in This Research
Python 3.12+	Core programming language	Data processing, model development, simulation scripting
OSMnx	Road network acquisition	Download and analyse Colombo street network from OpenStreetMap [11]
SUMO 1.20+	Traffic simulation	Microscopic simulation for model validation [5][6]
SciPy	Optimization	Weight optimization (SLSQP), KDE, statistical tests
scikit-learn	Machine learning	DBSCAN clustering [8], Random Forest, Gradient Boosting comparison
pandas / NumPy	Data manipulation	Incident data processing, CSV handling, numerical computation
matplotlib / seaborn	Visualization	Charts, scatter plots, analysis figures
Folium	Interactive maps	Hotspot map generation
Next.js 16	Dashboard frontend	React-based web application with App Router
Leaflet.js	Map rendering	Interactive map with markers, heatmap, and cluster overlays
Tailwind CSS	Styling	Utility-first CSS framework for dashboard UI

Git / GitHub	Version control	Private repository for code and data management

Ethical Considerations

This research uses publicly available road network data from OpenStreetMap and synthetically generated incident data. No personal or identifiable data is collected or processed. The platform design incorporates PDPA compliance principles including purpose limitation, data minimization, and user consent mechanisms. Traffic simulation data is generated computationally and does not involve human subjects.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The Kaduna.lk platform follows a modular architecture where four independent components interact through well-defined API interfaces. The Geo-Intelligence engine serves as the spatial awareness layer, providing traffic intelligence to the dispatch system and visualization to end users.

Overall Platform Architecture

The platform comprises four core components, each led by one team member:

Member	Component	Scope
Janukshan S (IT22635266)	Dispatch Optimization & Smart Recovery	Automated service provider allocation using optimization algorithms
Asath M M (IT22633422)	Geo-Intelligence & Traffic Impact Analysis	Incident impact scoring, congestion prediction, hotspot analysis, visualization dashboard
Herath D M S T (IT22639776)	Predictive Maintenance for Service Management	AI/IoT-based vehicle failure prediction and maintenance alerts
De Silva R K D H (IT22001252)	Compliance, Integration & Stakeholder Engagement	PDPA compliance, stakeholder integration, governance policies

Geo-Intelligence Engine Architecture

The Geo-Intelligence engine is structured in three layers:

- Data Layer: Road network database (downloaded from OpenStreetMap via OSMnx), traffic volume estimates (HCM-based capacity tables), and incident data (synthetic records with GPS coordinates, type, time, and road attributes).

- Analysis Engine: Impact Scoring Model (5-factor weighted formula), Congestion Prediction Model (LWR shockwave theory), Hotspot Analysis (KDE + DBSCAN), and What-If Simulator (parameter-adjustable scenario prediction).
- Output Layer: Priority Signal API (JSON responses to dispatch engine), Visualization Dashboard (Next.js + Leaflet.js interactive map), and Reports/Analytics (hotspot maps, impact statistics, comparison tables).

Component Interactions and Data Flow

When an incident is reported, the Geo-Intelligence engine processes it through the following pipeline: (1) GPS coordinates are matched to the nearest road segment in the network database, (2) road attributes (type, lanes, speed limit) and traffic state (estimated volume based on time of day) are retrieved, (3) the impact scoring model calculates a severity score and priority level, (4) the congestion prediction model estimates queue length, vehicle-hours lost, and recovery time, (5) results are sent to the dispatch engine as a priority signal and displayed on the dashboard.

Component Boundary Matrix

Dimension	Janukshan (Dispatch)	Asath (Geo-Intelligence)	Herath (Maintenance)	De Silva (Compliance)
Core Question	Which provider should respond?	How bad will this incident be for traffic?	Can we predict vehicle failures?	Is the platform legally compliant?
Research Domain	Operations Research, VRP	Transportation Engineering, GIS, Spatial Analysis	Machine Learning, IoT	Data Protection Law, UX
Algorithm/Method	Greedy, Heuristic, Genetic Algorithm	Weighted scoring, Shockwave theory, DBSCAN, SUMO	Random Forest, Neural Networks	DPIA framework, Consent workflows
Output	Provider assignment + ETA	Impact score, congestion prediction, hotspot maps	Failure probability, maintenance alerts	Privacy policy, compliance checklist
Evaluation	Response time, travel distance	Scoring accuracy, prediction error, VHL reduction	Prediction accuracy (F1)	PDPA compliance %

The Geo-Intelligence engine provides the dispatch system with impact scores and priority levels via a REST API. The dispatch system returns dispatch timestamps and provider ETAs, which are used to update congestion predictions and dashboard displays. These are API-level interactions—data passes through defined interfaces, and neither component modifies the other's logic.

Scalability and Security Considerations

The architecture is designed for future scalability. The scoring model operates as a stateless function that can be deployed behind a load balancer for horizontal scaling. The road network database is read-only during operation and can be cached in memory for sub-millisecond lookups. The dashboard uses static site generation (Next.js export) for deployment on CDN infrastructure, ensuring low-latency access globally.

Security measures include PDPA-compliant data handling with purpose limitation (location data used only for scoring and visualization), data minimization (no personal identifiers stored beyond session), and HTTPS encryption for all API communications. The REST API will implement rate limiting and API key authentication to prevent abuse. Synthetic incident data used during development contains no real personal information.

5. USER REQUIREMENTS

Stakeholder Identification

Stakeholder	Primary Need
Stranded Drivers	Request emergency assistance, view estimated response time and traffic impact
Roadside Assistance Providers	Receive prioritized dispatch notifications with impact context
Traffic Authorities	Monitor incident-induced congestion, identify hotspot zones, plan infrastructure
Platform Operators	Manage incidents, monitor system performance, configure scoring parameters
Researchers	Access what-if simulation tool for scenario analysis and traffic planning

Functional Requirements

ID	Requirement
FR-01	The system shall calculate an impact score (1–10) for any reported incident within 500ms
FR-02	The system shall classify incidents into CRITICAL, HIGH, MEDIUM, or LOW priority
FR-03	The system shall predict queue length, vehicle-hours lost, and recovery time for each incident
FR-04	The system shall display incidents on an interactive map with color-coded severity markers
FR-05	The system shall provide a heatmap overlay showing incident density weighted by impact score
FR-06	The system shall identify and display hotspot zones using spatial clustering
FR-07	The system shall provide a what-if simulator for hypothetical incident analysis
FR-08	The system shall expose a REST API for integration with the dispatch engine
FR-09	The system shall support filtering incidents by priority level and road type
FR-10	The system shall display detailed score breakdown and factor analysis for each incident

Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Scoring API response time < 500ms
NFR-02	Performance	Dashboard load time < 3 seconds
NFR-03	Usability	Map rendering smooth (>30 FPS) with 50+ simultaneous incidents
NFR-04	Security	PDPA-compliant data handling with purpose limitation and consent
NFR-05	Reliability	Scoring model accuracy: Pearson r > 0.75 against simulation ground truth
NFR-06	Scalability	Architecture supports future integration with real-time traffic data APIs
NFR-07	Maintainability	Modular codebase with clear separation between data, analysis, and presentation layers

6. COMMERCIALIZATION PLAN

- **Target Market:** Daily commuters in urban Sri Lanka (Colombo, Kandy, Galle metropolitan areas), commercial fleet operators, insurance companies offering roadside assistance, and traffic management authorities seeking data-driven congestion intelligence.
- **Value Proposition:** The first traffic intelligence platform for Sri Lankan roads that transforms reactive roadside assistance into proactive, impact-aware emergency response. Unlike existing manual systems, Kaduna.lk's geo-intelligence engine tells providers not just WHERE to go, but WHY it matters—enabling smarter resource allocation that reduces both individual wait times and network-wide congestion.
- **Revenue Model:** Freemium model for individual drivers (free basic assistance requests, premium for priority dispatch and real-time congestion alerts). B2B subscription for fleet operators and insurance companies (API access to scoring and analytics). Government licensing for traffic authority dashboard and hotspot intelligence.
- **Cost Estimation:** Development costs are primarily human effort (4 team members × 10 months). Infrastructure costs include cloud hosting for the dashboard and API, Google Maps API usage for traffic data, and domain registration. Total estimated development cost: LKR 200,000–300,000.
- **Pricing Strategy:** Free tier for basic users; LKR 500/month for premium individual users; LKR 25,000–50,000/month for fleet operators (based on fleet size); custom pricing for government and institutional clients.
- **Competitive Advantage:** First-mover advantage in Sri Lankan market; proprietary impact scoring model calibrated for local road conditions; validated against traffic simulation; integrated platform approach (not standalone traffic app).
- **IP Considerations:** The impact scoring methodology and composite risk scoring algorithm may be eligible for intellectual property protection. The research findings will be published as academic contributions. The software codebase will be maintained as a private repository with potential for open-source release of non-proprietary components.

7. BUDGET AND JUSTIFICATION

Category	Item	Estimated Cost	Justification
Hardware	Development laptops (existing)	LKR 0	Team members use personal laptops
Software	SUMO traffic simulator	LKR 0	Open-source (GPL v2)
Software	Python + libraries (OSMnx, scikit-learn, etc.)	LKR 0	Open-source
Software	Next.js, Leaflet.js, Tailwind CSS	LKR 0	Open-source
Cloud	Vercel hosting (dashboard deployment)	LKR 0–15,000	Free tier for development; paid for production
Cloud	Google Maps API (traffic data)	LKR 10,000–20,000	Free tier covers ~28,000 requests/month; additional for heavy usage
Cloud	Domain registration (kaduna.lk)	LKR 5,000	Annual domain registration
Data	OpenStreetMap road network data	LKR 0	Freely available via OSMnx
Human Effort	4 team members × 10 months	LKR 0 (academic)	Final-year research project
Printing	Thesis printing and binding	LKR 10,000–15,000	Final submission copies
Miscellaneous	Internet, transportation, stationery	LKR 15,000–20,000	Ongoing project expenses
	TOTAL ESTIMATED	LKR 40,000–75,000	

8. WORK BREAKDOWN STRUCTURE (WBS)

WP	Task	Responsible	Timeline	Deliverables
WP1	Data Collection & Preparation	Asath	Month 1–2	Road network download, traffic estimation, synthetic incident generation
WP2	Impact Scoring Model	Asath	Month 3–4	Factor definition, formula implementation, initial weight assignment, API development
WP3	SUMO Simulation	Asath	Month 4–5	Network setup, 90 scenario runs, congestion metric collection, validation
WP4	Model Refinement	Asath	Month 5–6	Weight optimization (SLSQP), ML comparison, sensitivity analysis
WP5	Hotspot Analysis	Asath	Month 5–6	KDE density surface, DBSCAN clustering, correlation analysis, composite risk scoring
WP6	Dashboard Development	Asath	Month 7–8	Next.js frontend, Leaflet map, statistics panel, what-if simulator
WP7	Integration & Testing	All members	Month 8–9	API integration with dispatch, end-to-end testing, performance benchmarking
WP8	Evaluation & Documentation	Asath	Month 9–10	Comprehensive evaluation, thesis writing, presentation preparation

The workload is balanced across the team, with each member leading their respective component while collaborating on integration (WP7). Asath's geo-intelligence work proceeds independently through WP1–WP6, with integration points in WP7 where the scoring API connects to Janukshan's dispatch engine.

9. GANTT CHART

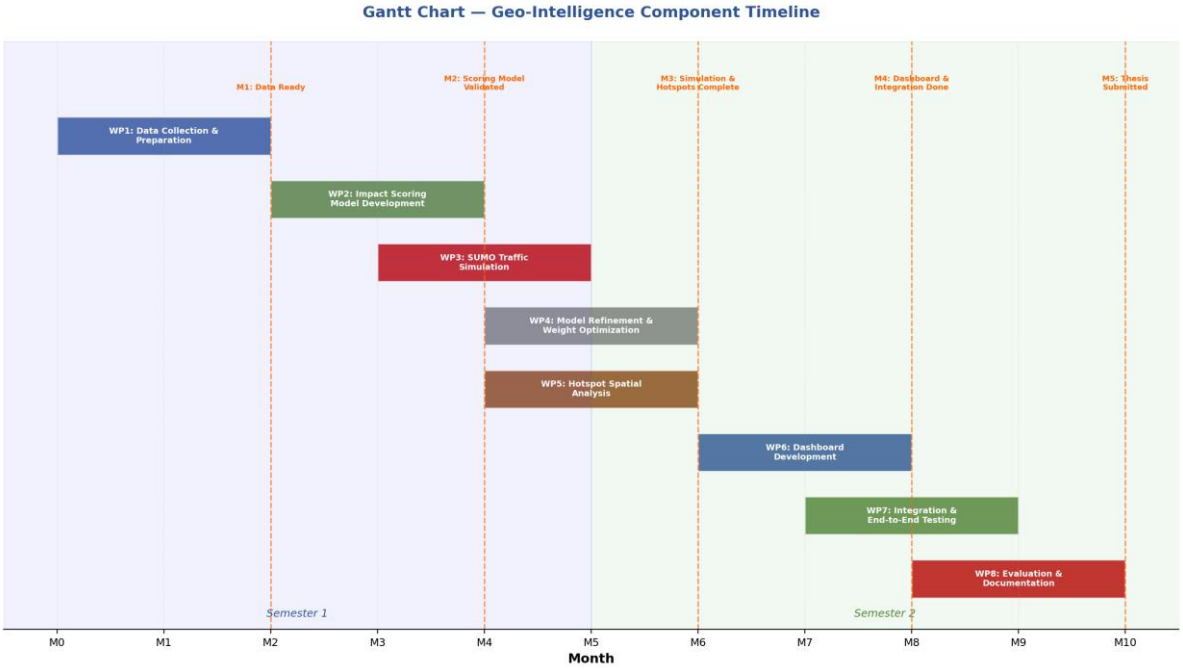


Figure: Gantt Chart — Geo-Intelligence Component Timeline

Timeline Summary:

Period	Phase	Key Deliverables
Month 1–2	Data Collection & Preparation	Road network database, processed datasets, study area definition
Month 3–4	Model Development & Calibration	Working scoring function, initial API, calibrated SUMO model
Month 5–6	Simulation, Refinement & Hotspot Analysis	Simulation results, optimized weights (target: r > 0.75), hotspot maps
Month 7–8	Dashboard & Integration	Working dashboard prototype, integrated system, what-if tool
Month 9–10	Evaluation & Documentation	Evaluation report, thesis draft, presentation slides

Key Milestones:

- Month 2: Road network and incident dataset ready
- Month 4: Impact scoring model validated (initial correlation)
- Month 6: SUMO validation complete (target: r > 0.75), hotspot analysis done
- Month 8: Dashboard deployed, API integrated with dispatch
- Month 10: Final thesis and presentation submitted

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APPENDICES

Appendix A: Impact Scoring Formula Derivation

The proposed scoring formula with initial weights:

$$Impact_Score = (w_1 \times CLF + w_2 \times TVF + w_3 \times TF + w_4 \times LF + w_5 \times ISF) \times 10$$

Where CLF = lanes_blocked / total_lanes, TVF = current_volume / road_capacity, TF = time-of-day multiplier (0.1–1.0), LF = road classification multiplier (0.2–1.0), ISF = incident type multiplier (0.2–1.0).

Initial weights: w_1=0.25, w_2=0.25, w_3=0.20, w_4=0.15, w_5=0.15 (to be optimized via SUMO).

Priority classification: CRITICAL (8.0–10.0), HIGH (5.0–7.9), MEDIUM (3.0–4.9), LOW (1.0–2.9).

Appendix B: Planned SUMO Simulation Configuration

Planned simulation parameters: Synthetic Road corridors for each road type, lane closure simulation using variableSpeedSign or rerouter elements, edge-level data collection at 60-second intervals, approximately 90 scenarios across 5 road types × 2 lane configs × 3 volume levels × 3 durations.

Appendix C: Sequence Diagram

The following sequence diagram illustrates the interaction flow when a vehicle incident is reported through the Kaduna.lk platform, showing how the Geo-Intelligence engine processes the incident and communicates with other system components.

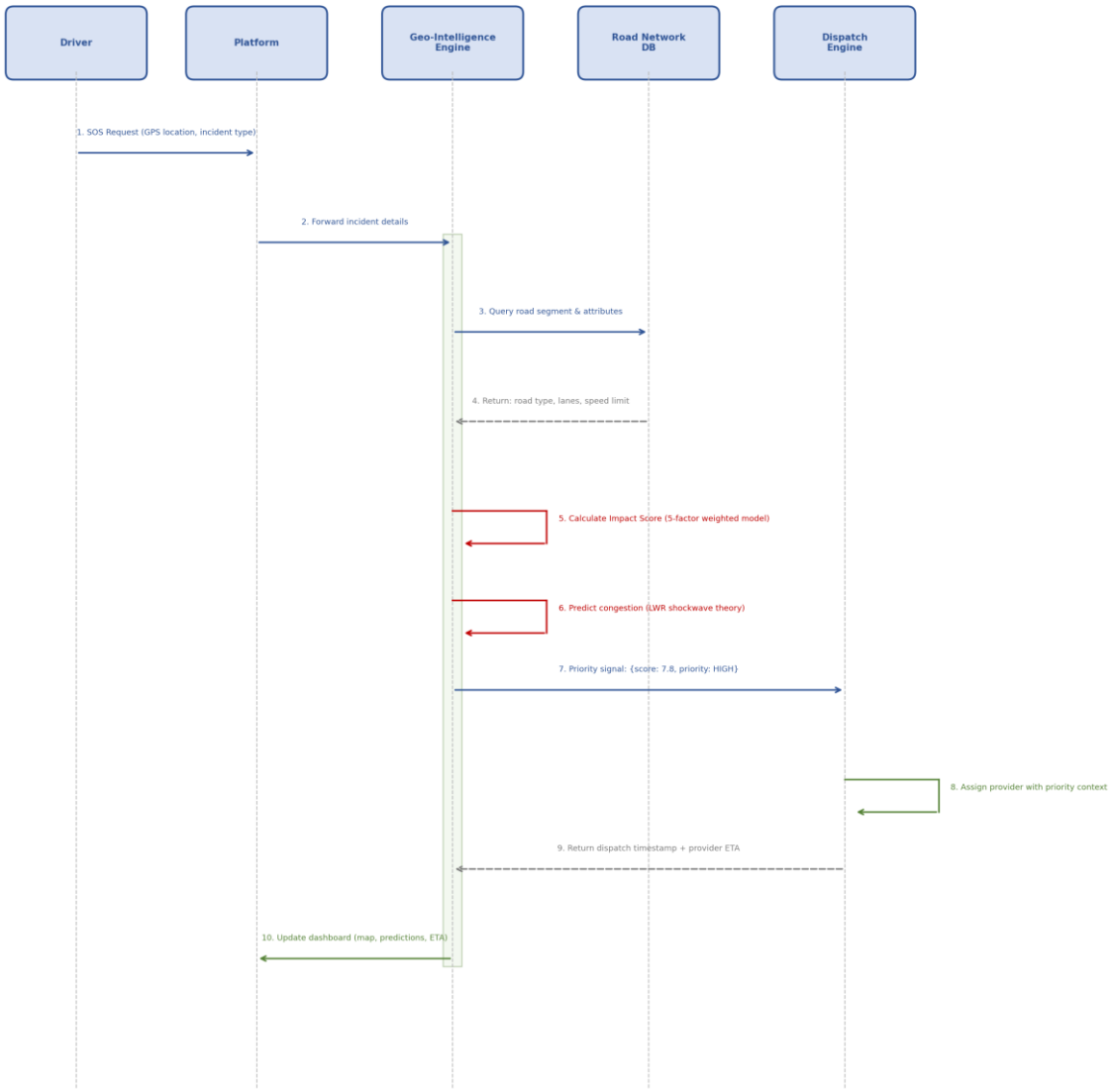


Figure 1: Sequence Diagram — Incident Scoring and Dispatch Flow

Figure: Sequence Diagram — Incident Scoring and Dispatch Flow

Appendix D: Risk Analysis

Risk	Likelihood	Impact	Mitigation
Real traffic data unavailable	High	Medium	Use Google Maps API estimates + synthetic data; document limitations
Real incident data unavailable	High	Medium	Generate synthetic dataset based on literature; validate with news reports
SUMO simulation too complex	Medium	High	Start with simple corridors; scale gradually; use default calibration
Impact scoring model low accuracy	Medium	Medium	Iterate on weights; try ML if formula insufficient; document findings
Integration with dispatch delayed	Low	Medium	Design scoring API independently; use mock dispatch data
Google Maps API costs exceed budget	Medium	Low	Use free tier; cache results; supplement with OSM data
Scope creep	Medium	High	Stick to 5 defined features; prioritize scoring model and hotspot analysis